Lesson Aim

Identify suitable climate and soil conditions for vineyard site establishment.

Introduction

The climate and soil type of a vineyard have a major impact on its success. In particular the mesoclimate (the overall climate) of a local geographic region can be quite influential on grape production and quality. While most climatic and soil factors can be manipulated to a small extent, the original conditions will dictate how a viticulturist will manage their vineyard, from initial plant selection and establishment right through to annual decisions regarding fertilising, irrigation, and harvest timing.

Four major factors need to be considered when selecting a suitable environment for growing grapes:

- Temperature and its variables
- The amount of sunlight available to the vine
- Rainfall and its seasonal distribution
- The soil in which the vine grows

Temperature

Suitable temperature is vital in enabling the vine to grow long enough to complete its annual growth cycle. Annual maximum and minimum temperatures are relevant, as is the annual temperature distribution (i.e. how long the growing season is). Varieties differ in their temperature tolerances, but grapes are generally produced in areas with a mean annual temperature of 10-20°C. Very little growth occurs below 10°C. Grapes require winter temperatures cool enough to induce vine dormancy, and a growing season long enough to allow fruit development.

Frosts of -4°C and below will damage leaves, kill young, tender shoots, plus damage flower clusters and, hence, prevent fruit set. Frost-free sites in the spring or autumn are quite valuable, otherwise frost protection measures may be required. Artificially elevating vineyard temperatures above potentially damaging levels can be achieved in a number of ways. In susceptible grape growing areas of the US, large fans are installed with the goal of mixing warm and cold air layers to avoid the ill-effects of an inversion layer. Areas in France and Italy that are subject to frost settling on tender vines have traditionally used oil-burning frost-pots to raise temperatures. Such devices need to be manned overnight and in the dangerous dawn hours.

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As with many potential pitfalls in grape growing, appropriate variety selection can alleviate, or minimise, problems. For example, growers in the UK living in areas that commonly receive severe frosts would be advised to select grapes varieties with late bud burst. The later succulent buds emerge, the more chance of escaping damage.

Temperature variables during ripening can affect the balance of sugars and pH (acidity) within the fruit. These are important factors in determining a grape's suitability for different uses. Vineyards in warmer climates usually produce table grapes, whereas wine grapes are usually produced by vineyards in cooler areas. Different cultivation methods can also be used to improve grape quality in various temperatures.

Typically different sites in the same locality can vary a great deal in their climatic characteristics and suitability for growing grapes.

- A WARM SITE has an aspect that faces the midday sun. It will miss early and late season frosts which affect other parts of the locality because cold air will fall into the valleys below.
- Where a slope facing the midday sun is at a higher altitude, the advantages found normally are counteracted by the cold which comes from being at a higher altitude.
- A COLD SITE has an aspect facing away from the midday sun. It heats up less in summer because it gets less direct sun. Being higher up the slopes it still may miss early and late frosts which settle in the valleys below.
- A VERY COLD SITE is a low spot in a valley which gets less direct sun than other sites and collects frosts earlier and later in the season than other places.
- AN AVERAGE SITE is flat. Frosts do not drain away as easily as on sloping ground. Sun is collected in summer or winter at an average rate. If on top of a hill it will be warmer than part way down a slope.

NOTE: Prevailing winds, shade from trees and buildings etc. can also affect the climatic characteristics of a site.

Temperature Calculations – Latitude-Temperature Index and Degree Days

An index of temperatures can be calculated to assess the suitability of a district for viticulture. These indexes allow comparison between different localities within the region. Various methods are used, utilising data such as monthly temperatures, latitude, or the mean temperature of the warmest month to estimate growth potential. Two of these methods are 'Degree Days' and 'Latitude-Temperature Index'.

The Degree Days calculation produces a temperature index based on the growth potential in a given month. The mean monthly temperature is obtained, and 10° is subtracted, since very little growth occurs below 10°C. The resulting figure is multiplied by the number of days in the month, to derive a 'degree days' index figure.

For example:

Month of June mean temperature: 18°C minus 10°C equals 8°C

8°C multiplied by 30 days equals 240°C.

Therefore, the ‘degree days’ for the region in this example is 240°C. This figure can be compared with other ‘degree days’ figures for other areas within the region, to compare available temperature potential.

By calculating monthly heat units from 20 years’ climatic data, a good basis for determining the suitability of various sites in a geographic area with similar climatic patterns can be obtained.

The Latitude-Temperature Index (LTI) is another method of evaluating the climatic suitability of a region for viticulture. This index takes account of the impact of latitude on the growing season. The LTI is calculated by subtracting the latitude of the region from the figure 60, and multiplying this figure by the mean temperature of the warmest month in the region.

For example:

Mean temperature of warmest month: 18°C multiplied by (60 minus 30° South Latitude) equals 540°C.

Therefore this region has an LTI of 540°C.

Sunlight

Sunlight is necessary for plant growth, since plants utilise sunlight to derive their energy from photosynthesis. Sunlight can also have an effect on other aspects of viticulture. Sunshine is particularly critical at the stage when fruit sets. Cloudy weather at this time is unfavourable because it decreases the production of sugar, leaving insufficient to supply the rapid growth involved in fruit setting. Sunlight can also play a part in disease reduction, since many diseases of grapes thrive in moist and poorly lit conditions.

Fruit that is exposed to too much direct sunlight can suffer damage. Good site selection can alleviate problems associated with poor sunlight exposure. A warm site catching more sun due to the ‘lie of the land’, or slope, can be advantageous. Such an aspect will serve to miss damaging frosts since cold air will drain to low-lying areas. Too high an elevation, however, will lead to its own problems with overall increased cold temperatures.

Rainfall

Grapes can survive in areas with very low rainfall. Although vines can grow well in high rainfall areas, vineyard management can be more difficult, especially where soils are poorly draining, or where rainfall occurs during fruit set. The timing of rainfall on the vineyard greatly affects production and quality. Wet soils are less of a problem in winter, while the vines are dormant, although prolonged and repeated water logging will result in vine damage. Copious rainfall events in the weeks leading up to harvest can result in continued, vigorous vegetative growth. Such an event will compromise the quality of the fruit. Only modest sugar content and flavour intensity will develop. Excessive rainfall can be a real danger with ripening grapes on the vine. The possibility of split berries and disease occurrence is potentially disastrous.

Of course, as with many crops, insufficient rainfall to sustain the plant will affect yields. In areas where there is such a possibility it is advisable to have an irrigation plan. The grower should not wait until the vines display clear signs of water stress before applying irrigation. Special, permanent irrigation regimes may become essential in areas of consistent low rainfall. Many of the world’s best viticultural regions are located in areas that receive less than 750mm of rain per year. These areas have much lower incidences of disease and have greater flexibility in the timing of the harvest, i.e. there is less probability of fruit rot at harvest time.
The Rhine and Mosel regions receive adequate rain in the spring to permit good vine growth without irrigation but little rain in the autumn to interfere with harvest. Regions with high autumn rainfalls can be problematic.

Soil

Soils vary from region to region, and even from locality to locality within a region. The relative amounts, and the quality, of each of the components can vary in different soils, affecting their horticultural potential and ability to support plant growth.

The majority of vine roots are located in the top one to one and a half metres of soil, but deeper roots frequently go down four, or even up to ten metres, if the soil allows this growth. These deep roots are said to influence grape quality through contact with subsurface soil types and rock strata. However, this deep root system makes vines susceptible to wet soil. The physical characteristics of soil should allow deep drainage, whilst retaining sufficient moisture to allow plant survival.

Trace elements extracted from the various layers of soil contribute greatly to the final flavour of the grape, and the wine made from those grapes. It is often poor soils which make fine wines while richer alluvial soils tend to make more ordinary wines.

The physical characteristics as well as the chemical components have an effect on the grape flavour. French Champagne is best made from grapes grown on solid chalk soils. The alkalinity of these soils makes the wine more acid, a necessary characteristic in champagne. The great clarets of Bordeaux are made from vines that grow on poor gravely soils. Fine delicate moselles from Germany grow on broken flakes of slate. Some of the finest ports in the world are grown in the Douro hills on heavy granitic and schistous rock.

Soil Types and Wine Regions

Different varieties of vines are frequently reputed to produce better grapes on particular types of soils. There are particular soils, such as the schistose slate base of Germany's Mosel region, the famous chalk and quartz areas of France (e.g. Chateau Mouton Rothschild), and the red 'terra rossa' soils of the Coonawarra region of South Australia that are quite distinctive. Gravely well-drained soils of low fertility are commonly found in the USA's Napa Valley (along with a diversity of other soil types) and are good for producing a number of fine varieties, e.g. Chardonnay and Riesling. Calcareous soils are traditionally associated with chardonnay, whereas cabernet sauvignon is said to benefit from granite soils.

Understanding Soils

Understanding soils is vital to successful viticulture. The success or otherwise with which vines may be able to grow in soil will greatly depend upon the soil's physical and chemical properties.

Soil is important to the plant in providing the following:

- Nutrition: The plant derives its food from nutrients in the soil.
- Support: The soil holds the plant firm and stops it falling over.
- Water and air: Roots absorb both water and air. The soil must contain both. A soil with too much air leaves the plant starved for water. A soil with too much water leaves the plant starved for air.

Different soils have different characteristic with respect to the above factors. For example, a sandy soil provides less support than a clay soil. A clay soil provides less air, but has a greater capacity to hold water than sand. An organic soil has a good ability to hold water, but doesn't always provide good support.

Physical Characteristics of Soil

Soil is made up of the following components:

- Mineral particles – sand, silt, clay, and other minerals
- Organic matter – humus and the remains of plants and animals
- Water and dissolved nutrients
- Air supplying oxygen to the plants
- Living organisms (worms, fungi, insects, micro-organisms etc.)

These things affect the soil's ability to grow plants. It is possible to grow some plants in soils without living organisms, organic matter or mineral particles, but plant roots must have air, water and nutrients. Generally, however, you will require some amount of each of the above components to get the best growth from your plant.

There are basically four component particles in soil:

- Gravel: particles larger than 2mm
- Sand: particles between 0.02 to 2mm in diameter
- Silt: particles between 0.02 and 0.002mm in diameter
- Colloids: particles less than 0.002mm in diameter

Soil Texture

Soil texture is the relative percentage of sand, silt and clay particles in a soil. A soil's texture affects the size of soil pores (air spaces between particles), absorption of water and nutrients, movement of air and water, and soil erosion.

A "soil texture triangle" can be used to identify and name texture, e.g. fine – coarse. The three points of the triangle are clay, sand and loam (silt). Most soils have a ratio of all three of these basic components.

Soil Structure

Soil structure is the arrangement of particles in the soil. The arrangement of particles determines whether a soil has adequate air space between the particles to supply air and water to the plant roots.

Colloids

The term 'colloid' describes the ability of soil particles to remain suspended in water for long periods of time. Colloids are very small mineral and organic particles – too small to see without an electron microscope. They function in soil by holding and supplying nutrients to plant roots. Due to their very small size, they represent an enormous surface area on which nutrients may be held.

Soils with a high proportion of dispersing clay colloids tend to have poor structure (eg. leached clay soils which are high in sodium). Air spaces (pores) between particles are reduced, potentially resulting in the following:

- Poor drainage
- Increased soil compaction
- Reduced oxygen available to roots
- Poor root penetration
- Poor fertiliser uptake
- Increased tendency for soil erosion

Improving a soil's structure has the effect of flocculating the colloids; in other words, clumping or aggregating the particles together to form structured peds (see below). Adding organic matter significantly improves a soil's ability to form peds as it acts as a natural cementing agent, binding the mineral colloids and stabilising the peds. This improves the balance between drainage and water retention, and also gives plant roots access to the greatest possible quantity of nutrients on the colloids.

Soil peds

As described above, the structure of a soil affects the ability of water and air to percolate into the soil, and can inhibit the penetration ability of plant roots. Soil can be structured so that the various soil particles aggregate into larger components called 'peds'.

Different soils have different ped structures - blocky, angular, prismatic or columnar structures (these terms relate to the appearance of the peds).

Structureless soils have no noticeable peds. They may be massive (a single consolidated mass of soil that does not readily break up into peds) or single grained (individual soil particles that do not cohere to one another to form peds). Heavy clay soils tend to be massive; very sandy soils are unconsolidated.

In strongly structured soils, most of the soil mass is visible as peds and these peds can be handled without their breaking.

A simple field test to determine soil structure is to take a lump of soil and drop it from a height of about a metre onto a hard surface. An examination of how the soil crumbles on impact will reveal the size and shape of individual peds. The degree to which a soil maintains a massive structure, or crumbles into individual grains, can also be observed.

Characteristics of Sand, Clay and Loam Soils

The nature of a soil depends on its texture and structure. Soils with a high proportion of sand particles have large air spaces between the particles, hence drain rapidly. These are usually better for propagating seeds or cuttings, but need to be kept well-watered. Sandy soils are generally low in organic matter and nutrients.

Clay soils have a high proportion of small particles, with small air spaces between the particles. Water is absorbed slowly and percolates (drains) slowly through to the lower layers. Clay soils tend to remain waterlogged during prolonged periods of rain or irrigation, they are prone to compaction while wet, and they can be difficult to cultivate. As they dry out, they may become cloddy and cracked. They generally have moderate levels of organic matter and good levels of nutrients.

Loam soils vary widely in their composition and structure. The best loam soils often contain a high proportion of silt particles (silty loams). They contain good levels of nutrients, they have moderate to good amounts of organic matter, they drain freely and hold sufficient water for plant roots. They are easy to cultivate and are able to support a wide variety of plant types, including many of our important crop varieties. However, not all loams provide these ideal conditions – some have a hard crust which repels water, others are low in nutrients.

Soil Profile

Soil consists of several layers, generally referred to as topsoil, subsoil, bedrock etc. These layers are collectively referred to as the soil profile. Soil characteristics such as structure, texture, colour, and nutrient content can vary from layer to layer.

This can be significant in viticulture since grape vines have such deep root systems. Impermeable sub-surface layers can limit drainage. On the other hand, particular sub-surface soils or rock may improve grape quality.

Soil Temperature

The rate of absorption of water and nutrients is affected by the temperature of the soil. Too much heat or cold will slow the whole metabolism down. Soil temperature is not always the same as atmospheric temperature. Mulching a plant or adding organic matter to the soil will even out (or lessen) the fluctuations in soil temperature. As with most organisms, plant roots will grow within a particular range of tolerance which will vary from one species to another.

Chemical Characteristics of Soils

Soils vary enormously in their chemistry and the soil supports many chemical reactions and interactions. Factors such as nutrient balance, acidity or alkalinity (pH) and salinity can be quantified using field or laboratory tests. Other factors such as cation exchange capacity (CEC – a soil's ability to retain nutrients in a form available to plants), organic matter levels and biological activity can also be quantified. Soil pH and nutrient levels are the most commonly used chemical characteristics to evaluate the suitability of soil for vine growth.

pH

The level of soil acidity or alkalinity is determined by the concentration of hydrogen ions in the soil. The pH scale ranges from 0 to 14, with 0 being the most acid, 14 the most alkaline, and 7 neutral. Soils usually range from a pH of around 4 to a pH of around 9. Soil pH affects nutrient availability and plant growth. Simple field test kits can be obtained to test pH.

Grape vines grow best in a slightly acidic soil, with pH levels between 5.5 and 6.5. Acid soils with a pH of 4.5 or less can result in deficiencies of magnesium and potassium; alkaline soils (greater than 7.0) can limit a range of micronutrients.

pH levels should be adjusted before planting the vines as the deep-rooted nature of grapevines makes post-liming or acidification treatments much less effective.

Nutrient Levels

Plants rely on a variety of nutrients to grow, and levels vary in different soils. Nitrogen, phosphorus and potassium are of particular importance to plant growth and development, but other minerals play equally significant roles in plant health and produce quality. Advanced laboratory equipment and techniques are required for accurate nutrient testing. Testing for a full range of nutrients can be beneficial in fertilising decisions.

Carbon levels are particularly important in maintaining soil health. Long-term trends in the ratio between carbon levels and nitrogen levels (the ‘C:N ratio’), and other nutrient levels, can provide insight into the effects of production on the soil.

Most soils have ample supplies of calcium and magnesium; hence fertilisers used in horticulture are usually almost completely made up mostly of the other major nutrients (nitrogen, phosphorus and potassium).

Every nutrient has its purpose, and a deficiency or oversupply of even a minor nutrient can have a major effect on the plant. Deficiencies can be difficult to detect, but as time passes symptoms will appear. Signs are stunted growth, unhealthy leaves that may be mottled, stunted and dying off, distorted stems and undeveloped root systems. If a nutrient is easily dissolved, the older leaves will be affected first, otherwise the growing tips, i.e. the new leaves, will be affected.

An oversupply of nutrients may initially cause extra growth, but may then become toxic, and plant growth will be reduced. Deficiencies are not always a result of the nutrient not being present; it may be the nutrient is being held in some form which prevents the plant taking it up, for example it could be attached to an insoluble material (known as "immobilisation" or being "locked up"), or be affected by pH.

Nutrient requirements and fertiliser regimes for viticulture enterprises are discussed in detail in Lesson 7.

Soil Water Content

Soil water content varies according to the soil type, the soil’s chemical make-up, the region’s climate, and the season. The plant roots need water and air (or oxygen) just as much as nutrients. Any given soil has a potential range from saturated to totally dry. Saturation is the point where all soil pores are completely filled with water, and the soil will not absorb any more. Once a soil has drained completely, water is still held in small soil pores, and adhered to soil minerals and organic matter. This point is called ‘field capacity’. Following continued evapotranspiration, without irrigation or rainfall to provide additional water, soil moisture levels will deplete to the point where plants are permanently wilted – the ‘permanent wilting point’. The soil still holds water but it is bound so tightly to the soil mineral particles that plant roots cannot extract it. Continued drying of the soil would lead to totally dry soil, although this rarely happens in the field.

Soil texture has a strong influence over the soil water holding capacity. Sandy soils generally drain more readily, and hold less water. Clay soils hold more water but can be prone to water logging. Grapes are most susceptible to water logging during early spring and periods of active growth.

Water

All nutrients entering a plant (except carbon and oxygen) do so when dissolved in water. Water itself is also needed by the plant for metabolism, where it is important in the processes of respiration and photosynthesis.

Air

Soil air is richer in carbon dioxide and poorer in oxygen than atmospheric air. The amount of soil air depends on the size of pore spaces between soil particles. Soil air is necessary! Only a few plants can survive with very little air about the roots. Thus, waterlogging will damage plants by depriving the roots of oxygen, not because there is too much water.

Simple Soil Tests

Measuring pH

Excess acidity may create toxic soil conditions due to the increase in solubility of such elements as aluminium and manganese. Soils which are too basic may cause deficiencies of essential nutrients such as boron, iron, zinc and manganese. Because the availability of many micronutrients is influenced by soil acidity, it is essential that accurate pH measurements be made.

Probe Meters

Probe meters consist of a probe on top of which is mounted a small meter sensitive to electric currents. The probe is simply pushed into the wet medium and the pH read off on the metre. These instruments have been found to be very unreliable and their use is not recommended.

Colorimetric Methods

In these methods, the colour obtained when the soil is mixed with an indicator solution is compared with those on a colour chart to give the pH. Kits are available to carry out these measurements. They are cheap and reliable and give results accurate enough for most landscaping and garden situations.

pH Meters

Meters for reading pH consist of an electrode and an electronics system which provides readout on a dial or a digital display. pH meters are very delicate and relatively expensive. Care must be used in their operation and they must be calibrated with standard solutions.

The measurement obtained with a pH meter depends on the method of extraction of the soil solution:

- Saturated paste extracts are the most easily standardised method. A minimum amount of water is added to just flood the sample. The solution is then extracted and tested.
- Volume soil; volume water method. This is easier but the results can be influenced by the soil characteristics.

Measuring the Water Content of Soil

Water content is the percentage of water in a soil, measured in terms of volume or mass. To calculate the water content of a container filled with soil, follow the steps below:

1. Take a typical sample of soil to be tested.
2. Weigh a small clean heat resistant ceramic container (e.g. a coffee mug). Kitchen scales will suffice. This weight reading equals "M".
3. Place approximately 20gm of soil in the container. Weigh the container with the soil. This weight reading equals "W".
4. Place the container (with soil) in an oven at approx. 100 degrees Centigrade for about 15 hours. As soon as it is cool enough, weigh the soil and container again. This reading equals "S".
5. Calculate the water content using the formula below:

For organic soils:

$$\text{Water content} = \frac{W - S}{W - M} \times 100\%$$

For non-organic soils:

$$\text{Water content} = \frac{W - S}{S - M} \times 100\%$$

Important note: This method of calculating soil water content is used currently in laboratories. As a student of the school, you may wish to alter this method to suit your convenience.

As a home study course, the program of learning you are following is designed to give you an idea on procedures in viticulture and we fully appreciate that without the use of a laboratory and associated facilities, some methods and techniques may need to be altered. With this in mind, we suggest you adjust the length of drying time for this particular method i.e. you may wish to reduce this to 90-120 minutes in the oven. Once complete, you should still carry out the calculation and record the results. Depending on the moisture content of the soil sample to begin with, you will most probably find it is not completely dried – don't worry. The purpose is to give you an understanding of overall methods.

Additionally, you should also note that if you heat above 100°C you will burn the organic material within the soil, thus change the chemistry of sample and make the results invalid.

Testing Drainage

Drainage can be tested easily by observing the way in which water moves through soil which is placed in a pot and watered. However, when soil is disturbed by digging, its characteristics may change. Another way, to get a more reliable result, is to use an empty tin can. With both the top and bottom removed it forms a parallel-sided tube which can be pushed into the soil to remove a relatively undisturbed sample. Leave a little room at the top to hold water, add some to see how it drains and then saturate the soil and add some more water to the top. You will note slower drainage on saturated and clay soils.

Naming the Soil

Soils are usually named according to texture. Work through the following steps to classify the soil

1. Place a small quantity of soil in the palm of your hand and add just enough water to make it plastic.

Does it:

1. Stain your fingers	Yes or No?
2. Bind together	Yes or No?
3. Feel gritty	Yes or No?
4. Feel silky or sticky	Yes or No?
5. Make water cloudy	Yes or No?

2. The following soil types will give typical answers:

	1.	2.	3.	4.	5.
a. SAND	NO	NO	YES	NO	NO
b. SANDY LOAM	NO	YES	YES	NO	NO
c. LOAM (or SILT)	NO	YES	NO	NO	NO
d. CLAY LOAM	YES	YES	NO	NO	YES
e. CLAY	YES	YES	NO	YES	YES

3. You should also be able to distinguish by the amount of grittiness whether it is a coarse, medium or fine sand. You will also find varying grades of other soil types by how well they bind together, etc. For example, a clay will bind so tightly that it can be rolled into a ball and formed into shapes (just like potters clay).

4. Organic soils are ones which have a large proportion of organic matter (more than 25%). These are usually black or brown in colour and feel silky. It is possible to get organic types of all of the above soils. Also, these soil types may be distinguished from others by placing a small amount in a glass of water. If most of it floats on top, then it is an organic soil.

Problems With Soils

Soils are constantly affected by all that goes on around them, both by natural processes and, particularly in recent history, by human activity. Some of the more common human-induced changes include:

- Reduction in vegetation cover, opening up soils to increased rates of erosion.
- Loss of soil structure due to poor cultivation techniques, and the passage of heavy equipment, or regular traffic by hoofed animals (e.g. cows, horses).
- Reduction in soil fertility by not replacing nutrient losses from agricultural production.
- An increase in saline-affected soils due to vegetation clearance and poor irrigation practices.
- Waterlogging due to poor irrigation practices.
- Soil acidification through the extensive use of acidifying fertilisers, or the extensive cropping of plants that remove large amounts of calcium from the soil (e.g. lucerne).
- Pollution of some soils through the use of persistent or toxic agricultural chemicals.

When vegetation is cleared from a site, soil is exposed to sunlight and the eroding effects of wind and water. Soil aeration is increased and the rate of weathering increases. Erosion can be caused by increased runoff on sites that have been cleared as there are no plant roots to absorb the excess water.

Apart from erosion, the proportion of organic matter in the soil gradually decreases, through the action of microbes in the soil which use it as a source of energy - unless the new land use provides some replacement.

A number of major soil-related problems can occur including:

- Loss of soil fertility
- Erosion
- Salinity
- Soil compaction
- Soil acidification
- Build-up of dangerous chemicals

Erosion

Soil erosion, which is the movement of soil particles from one place to another by wind or water, is considered to be a major environmental problem. Erosion has been going on through most of earth's history and has produced river valleys and shaped hills and mountains. Such erosion is generally slow, but the action of humans has caused a rapid increase in the rate at which soil is eroded (i.e. a rate faster than natural weathering of bedrock can produce new soil). This has resulted in a loss of productive soil from crop and grazing land, as well as layers of infertile soils being deposited on formerly fertile crop lands, the formation of gullies, siltation of lakes and streams, and land slips.

Common human causes of erosion include:

- Poor agricultural practices such as ploughing soil to poor to support cultivated plants or ploughing soil in areas where rainfall is insufficient to support continuous plant growth.
- Exposing soil on slopes.
- Removal of forest vegetation.
- Overgrazing - removing protective layers of vegetation and surface litter.
- Altering the characteristics of streams, causing bank erosion.
- Causing increased peak water discharges (increased erosion power) due to changes in hydrological regimes, by such means as altering the efficiency of channels (channel straightening); reducing evapotranspiration losses as a consequence of vegetation removal; and by the creation of impervious surfaces such as roads and footpaths, preventing infiltration into the soil and causing increased runoff into streams.

The two basic types of erosion are:

- Water erosion, where soil particles are detached either by splash erosion (caused by raindrops), or by the effect of running water.
- Wind erosion, where wind can quickly remove soil particles, including nutrients and organic material from the soil surface (generally the most fertile part of the soil). Dislodged soil can be lost from a property, or may be deposited in places where it is a problem, such as against fences or walls, or on roads. Windblown particles can also damage plants (i.e. sandblasting effect).

Controlling Erosion

As erosion is caused by the effects of wind and water, control methods are generally aimed at modifying these effects. Some of the most common control methods are listed below:

- Prevention of soil detachment by the use of cover materials such as plants (i.e. trees, mulches, stubbles, crops).
- Crop production techniques (e.g. fertilising), to promote plant growth and hence surface cover.
- Strip cropping (strips of cereal alternated with strips of pasture or other crop) - hence no expanse of soil remains bare for a length of time.
- Ploughing to destroy rills and contour planting to create small dams across a field, to retard or impound water flow.
- Filling small gullies with mechanical equipment or conversion of the gully into a protected or grassed waterway.
- Terracing of slopes to reduce rates of runoff.
- Prevention of erosion in the first place by careful selection of land use practices.
- Conservation tillage methods.
- Armouring of channels with rocks, tyres, concrete, timber, etc., to prevent bank erosion.
- The use of wind breaks to modify wind action.
- Ploughing into clod sizes too big to be eroded, or ploughing into ridges.
- Avoiding long periods of fallow.
- Working organic matter into soil.

Salinity

High salt levels in soils reduce the ability of plants to grow or even to survive. This is can be caused by natural processes, but much occurs as a consequence of human action. Dryland salinity is caused by the discharge of saline groundwater, where it intersects the surface topography. This often occurs at the base of hills or in depressions within the hills or mountains themselves.

Wetland salinity occurs where irrigation practices have caused a rise in water-tables, bringing saline groundwater within reach of plant roots. This is common on lower slopes and plains and is particularly common on riverine plains. The wetland salinity problem is compounded by rises in groundwater flow due to dry land salinisation processes higher in the catchment.

Grape vines are considered to be moderately sensitive to salinity. Some rootstocks have greater salt tolerance than others, hence appropriate rootstock selection is important in saline-affected areas.

The presence of salinity is often indicated by:

- Yellowing of pasture (NB: waterlogging or drought can also cause this effect)
- Decline in crop plants, including yellowing, or burning of the tips of foliage
- Bare patches developing
- Reddish leaves on plantain
- Unusual species of weeds or grasses appearing (known as indicator plants)
- Salt deposits appearing at the soil surface (in more severe cases)

Some of the main methods used to control salinity are:

- Pumping to lower groundwater levels, with the groundwater being pumped to evaporation basins or drainage systems
- Careful irrigation practices to prevent a rise in, or to reduce groundwater levels
- 'Laser' grading to remove depressions and best utilise water on crop and grazing land
- Use of saline-resistant plant species
- Revegetation of 'recharge' areas and discharge sites
- Engineering methods designed to remove saline water from crop land
- Leaching suitable soils (e.g. raised crop beds)
- Identifying high recharge areas - these are often unproductive ridges. These areas are best planted with trees
- Applying gypsum to surface soils with high Sodium chloride (common salt) levels. The calcium and magnesium in the gypsum will displace the sodium ions from soil particles allowing them to be more readily leached by heavy irrigation or flood irrigation.

Soil Structural Decline

This causes a reduction in soil pore space, reducing the rate at which water can infiltrate and drain through the soil. It also reduces the available space for oxygen in the plant root zones, and makes it difficult for plant roots to penetrate through the soil. Some of the major consequences of soil structural decline are poor drainage, poor aeration, and hard pan surfaces which results in poor infiltration rates, and hence increased surface runoff.

Soil Acidification

This problem is becoming increasingly more common in cultivated soils. The pH of a soil can have major effects on plant growth, as various nutrients become unavailable for plant use at different pH levels.

Most plants prefer a slightly acid soil, however an increase in soil acidity to the levels being found in many areas of cultivated land renders that land unsuitable for many crops, or requires extensive amelioration works to be undertaken.

Chemical Residues

Although not as large a problem as some of the other types of soil degradation, the presence of chemical residues can be quite a problem on a local scale. These residues derive almost entirely from long-term accumulation after repeated use of pesticides, or of use of pesticides or other chemicals with long residual effects.